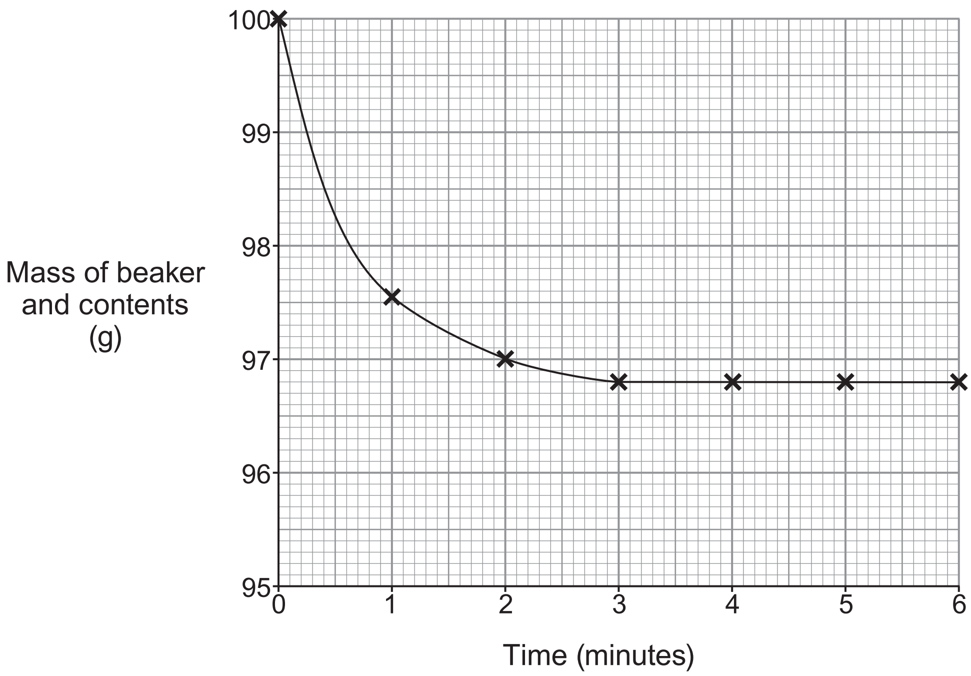
Examiner only

7. Powdered calcium carbonate was added to an excess of dilute hydrochloric acid in a beaker.

The mass of the beaker and its contents was recorded every minute for 6 minutes.

The graph shows the results.



The equation for the reaction is

$$CaCO_3 + 2HCl \longrightarrow CaCl_2 + H_2O + CO_2$$

(a) Describe how the mass of the beaker and its contents changes over the first minute. Give the reason for this change. [2]

....

....

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(b) (i) Use the graph to determine the time taken for the reaction to finish. [1]

....

(ii) When is the reaction at its fastest? Tick (✓) the correct box. [1]

from 0 – 0.5 minutes	☐
from 1 – 1.5 minutes	☐
from 2 – 2.5 minutes	☐
from 3 – 3.5 minutes	☐

(c) Use your graph to calculate the mean rate of the reaction during the **first two minutes**.

Use the following equation. [2]

$$\text{rate} = \frac{\text{decrease in mass of beaker and contents (g)}}{\text{time (minutes)}}$$

Rate = ... g / minute

(d) The experiment was repeated using the same mass of calcium carbonate but as a **lump** instead of a powder.

On the grid opposite, sketch the graph you would expect to obtain from this second experiment. [1]

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